HUJI Research Monitor

Healthcare · Weekly Digest · Week 28 · July 06, 2026

EXECUTIVE SUMMARY

This week's digest highlights a diverse portfolio of groundbreaking innovations from Hebrew University, ranging from transformative cell therapies to intelligent diagnostic platforms. Featured research includes pioneering CAR T-cell treatments for autoimmune diseases and stem cell-derived therapies for neurological disorders, addressing critical unmet medical needs. Additionally, we showcase advancements in precision oncology, high-efficiency drug discovery tools, and scalable medical devices poised to significantly impact global healthcare markets and offer compelling investment opportunities.

#1

B cell Maturation Antigen Targeted Chimeric Antigen Receptor T Cell Therapy for Refractory Systemic Lupus Erythematosus and Systemic Sclerosis.

37/50

Main Researcher: Chamutal Gur

[Immunology] [Clinical] [Synthetic Biology]

"Breakthrough CAR T-cell therapy HBI0101 shows promise for severe autoimmune diseases like lupus and systemic sclerosis."

WHY NOW

This represents a paradigm shift from oncology, tapping into a high-value chronic disease market with significant unmet needs for curative treatments. Early clinical data supports its disruptive potential.

COMMERCIAL ANGLE

Developing specialized CAR T-cell therapies for autoimmune indications represents a massive shift from oncology toward high-value chronic disease management markets.

https://europepmc.org/article/MED/42098533

#2

Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders.

36/50

Main Researcher: Tamir Ben-Hur

[Neuroscience] [Immunology] [Clinical] [Synthetic Biology]

"Off-the-shelf stem cell-derived astrocyte therapy offers new hope for remyelination in MS and demyelinating disorders."

WHY NOW

Demyelinating disorders like MS have vast unmet needs for repair strategies, and a scalable, clinical-grade cell therapy could revolutionize neurological treatment.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

https://pubmed.ncbi.nlm.nih.gov/42276062/

#3

The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms.

33/50

Main Researcher: Oren Zimhony

[Medical Device] [Materials] [Clinical]

"OxiLast™ antimicrobial surface coating significantly boosts hospital infection control, targeting multidrug-resistant bacteria."

WHY NOW

The global focus on healthcare-associated infections (HAIs) and antibiotic resistance creates an urgent market need for durable, scalable solutions in hospital environments.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

https://pubmed.ncbi.nlm.nih.gov/42374536/

#4

RNA dicing promotes the expression of an oncogenic JAK1 isoform.

33/50

Main Researcher: Yuval Malka

[Drug Discovery] [Diagnostics] [Genomics] [Immunology]

"RNA dicing discovery enables precision medicine for endometrial cancer, stratifying patients for existing JAK1 inhibitors."

WHY NOW

This diagnostic innovation expands the market for established therapies by identifying new patient populations, offering a less risky pathway to market than de novo drug discovery.

COMMERCIAL ANGLE

The discovery enables a new precision medicine approach by using RNA dicing profiles to stratify patients for existing JAK1 inhibitors like momelotinib.

https://pubmed.ncbi.nlm.nih.gov/41984590/

#5

Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model.

33/50

Main Researcher: Amir Haze

[Medical Device] [Clinical] [Synthetic Biology] [Proteomics]

"Recombinant human amelogenin protein, Remelix®, demonstrates powerful healing for cartilage and bone injuries."

WHY NOW

With an aging population and high prevalence of osteoarthritis, there's a massive and growing market for regenerative biologics that can repair damaged joints effectively.

COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

https://pubmed.ncbi.nlm.nih.gov/41925697/

#6

Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein.

29/50

Main Researcher: Rachel Nechushtai

[Drug Discovery] [Clinical] [Immunology]

"Novel peptide targets and inhibits aggressive primary and metastatic ovarian cancer, signaling a first-in-class oncology drug."

WHY NOW

Ovarian cancer remains a high-mortality disease with a critical need for new, targeted therapies, making this peptide a prime candidate for pharmaceutical licensing.

COMMERCIAL ANGLE

The peptide represents a high-potential therapeutic candidate for a first-in-class oncology drug, suitable for licensing to pharmaceutical companies focusing on precision cancer therapies or protein-protein interaction inhibitors.

https://pubmed.ncbi.nlm.nih.gov/42259134/

#7

Breath-Based Detection of Oral Diseases Using Sensors and Machine Learning.

28/50

Main Researcher: Yael Houri-Haddad

[Medical Device] [Diagnostics] [Software/AI]

"Breathalyzer for oral diseases, using AI and nanosensors, offers rapid, non-invasive diagnostic for dental clinics."

WHY NOW

The dental industry seeks objective, convenient diagnostic tools, and a non-invasive, AI-powered breathalyzer provides a significant upgrade over current subjective assessments.

COMMERCIAL ANGLE

Development of a non-invasive, rapid-screening handheld breathalyzer for dental clinics to provide objective diagnostics for periodontal health.

https://pubmed.ncbi.nlm.nih.gov/42379259/

#8

RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

Main Researcher: Ofer Mandelboim

[Immunology] [Drug Discovery] [Clinical]

"Bacterial protein RadD activates NK cells for enhanced antitumor cytotoxicity, paving way for novel immunotherapies."

WHY NOW

Natural killer cell-based immunotherapies are a rapidly expanding and highly promising area in oncology, with RadD offering a distinct, potentially powerful activation mechanism.

COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

https://europepmc.org/article/MED/42065370

#9

Aging-associated modulation of UFMylation impairs proteostasis in C. elegans.

27/50

Main Researcher: Ehud Cohen

[Drug Discovery] [Neuroscience] [Proteomics]

"Targeting UFMylation pathway in worms extends lifespan and clears toxic proteins, promising therapies for neurodegeneration."

WHY NOW

The immense societal burden and lack of disease-modifying therapies for Alzheimer's and Parkinson's drive urgent demand for novel drug targets, even at early discovery stages.

COMMERCIAL ANGLE

Targeting the UFMylation pathway offers a novel pharmacological strategy for developing disease-modifying therapies for neurodegenerative disorders like Alzheimer's and Parkinson's.

https://pubmed.ncbi.nlm.nih.gov/42056101/

#10

Targeting Temozolomide-Resistant Glioblastoma: Therapeutic Potential of Neuronal Nitric Oxide Synthase Inhibitor.

26/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Clinical]

"Nitric oxide inhibitor overcomes Temozolomide resistance in glioblastoma, offering new adjuvant therapy for deadly brain cancer."

WHY NOW

Glioblastoma remains notoriously difficult to treat, and a strategy to enhance existing chemotherapy effectiveness addresses a critical, high-value unmet medical need.

COMMERCIAL ANGLE

Development of a high-value adjuvant therapy to extend the clinical utility of existing standard-of-care chemotherapy for resistant brain tumors.

https://pubmed.ncbi.nlm.nih.gov/42374654/

#11

Strategic Template Filtering Accelerates Fragment-Based Peptide Docking.

26/50

Main Researcher: Ora Schueler-Furman

[Drug Discovery] [Software/AI] [Proteomics]

"AI-driven PatchMAN2 accelerates peptide docking, optimizing drug discovery platforms for pharmaceutical R&D.;"

WHY NOW

Pharmaceutical companies are heavily investing in AI/ML to reduce drug discovery timelines and costs, making this high-efficiency computational tool extremely valuable for licensing.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

https://pubmed.ncbi.nlm.nih.gov/42299509/

#12

Safety and Efficacy of Direct Selective Laser Trabeculoplasty in Patients With Ocular Hypertension or Open Angle Glaucoma.

26/50

Main Researcher: David Zadok

[Medical Device] [Clinical]

"Direct Selective Laser Trabeculoplasty (DSLT) proves safe and effective for glaucoma, expanding laser surgical options."

WHY NOW

Patients and clinicians are seeking alternatives to daily eye drops for glaucoma management, positioning laser-based procedures like DSLT for significant market adoption as a first-line treatment.

COMMERCIAL ANGLE

The findings support the commercialization and expanded clinical adoption of laser-based ophthalmic surgical devices as a first-line alternative to traditional glaucoma eye drops.

https://europepmc.org/article/MED/42148860